

An Infinite Pell Family for $z^2 + xy^2 + x^3 + 3 = 0$

Abstract

We prove, by an explicit elementary construction, that

$$z^2 + xy^2 + x^3 + 3 = 0$$

has infinitely many integer solutions. Fixing $x = -3$ reduces the equation to the Pell-type conic $z^2 - 3y^2 = 24$. A norm-preserving recurrence, obtained from the unit $2 + \sqrt{3}$, then generates an infinite strictly increasing sequence of solutions. We also explain in detail how this useful slice and its recurrence arise.

1 The explicit infinite family

Theorem 1. Define integer sequences $(y_n)_{n \geq 0}$ and $(z_n)_{n \geq 0}$ by

$$y_0 = 2, \quad z_0 = 6, \quad y_{n+1} = 2y_n + z_n, \quad z_{n+1} = 3y_n + 2z_n. \quad (1)$$

Then, for every $n \geq 0$ and every $\varepsilon, \delta \in \{\pm 1\}$,

$$(x, y, z) = (-3, \varepsilon y_n, \delta z_n) \quad (2)$$

is an integer solution of

$$z^2 + xy^2 + x^3 + 3 = 0. \quad (3)$$

Moreover, the solutions $(-3, y_n, z_n)$ are pairwise distinct. Consequently, (3) has infinitely many integer solutions.

Proof. Put

$$\mathcal{Q}(y, z) = z^2 - 3y^2.$$

A direct calculation gives the invariant identity

$$\begin{aligned} \mathcal{Q}(2y + z, 3y + 2z) &= (3y + 2z)^2 - 3(2y + z)^2 \\ &= z^2 - 3y^2 = \mathcal{Q}(y, z). \end{aligned}$$

Since

$$\mathcal{Q}(y_0, z_0) = 6^2 - 3 \cdot 2^2 = 24,$$

induction using (1) yields

$$z_n^2 - 3y_n^2 = 24 \quad (n \geq 0). \quad (4)$$

On taking $x = -3$, the left-hand side of (3) becomes

$$z_n^2 - 3y_n^2 - 27 + 3 = 24 - 24 = 0$$

after applying (4). Because only y^2 and z^2 occur in (3), the two signs in (2) may be chosen independently.

It remains only to verify infinitude. The initial values are positive, and (1) therefore implies inductively that $y_n, z_n > 0$ for every n . Hence

$$y_{n+1} = 2y_n + z_n > y_n,$$

so (y_n) is strictly increasing. Thus the triples $(-3, y_n, z_n)$ are pairwise distinct, proving the assertion. $\square$

The first positive solutions furnished by the theorem are

$$(-3, 2, 6), \quad (-3, 10, 18), \quad (-3, 38, 66), \quad (-3, 142, 246), \quad (-3, 530, 918), \dots$$

2 How the construction was found

The decisive step is to expose an indefinite quadratic form hidden inside the cubic equation. First, no solution can have $x \geq 0$: in that case z^2 , xy^2 , and x^3 are all nonnegative, so the left-hand side of (3) is at least 3. Every integer solution must therefore have $x < 0$.

Write $x = -a$ with $a \in \mathbb{Z}_{>0}$. Equation (3) is then exactly equivalent to

$$z^2 - ay^2 = a^3 - 3. \quad (5)$$

For each fixed a , the original cubic surface has been cut by the plane $x = -a$ into a Pell-type conic. The problem is now to choose a so that this conic has one integer point and also admits a nontrivial norm-preserving motion.

The first two choices fail for elementary congruence reasons. For $a = 1$, equation (5) reads

$$z^2 - y^2 = -2,$$

which is impossible modulo 4, since a difference of two squares is never congruent to 2 (mod 4). For $a = 2$, it reads

$$z^2 - 2y^2 = 5.$$

If y is even, this would give $z^2 \equiv 5 \pmod{8}$; if y is odd, it would give $z^2 \equiv 7 \pmod{8}$. Both are impossible because a square modulo 8 is 0, 1, or 4.

The next choice, $a = 3$, succeeds immediately:

$$z^2 - 3y^2 = 24, \quad (6)$$

and the small identity

$$6^2 - 3 \cdot 2^2 = 24$$

provides the seed $(y, z) = (2, 6)$. Thus the remaining task is not to find more points independently, but to find a transformation preserving the left-hand side of (6).

The general multiplicative identity

$$(rz + asy)^2 - a(sz + ry)^2 = (r^2 - as^2)(z^2 - ay^2) \quad (7)$$

is verified by expanding both sides: the mixed terms cancel, leaving $(r^2 - as^2)z^2 - a(r^2 - as^2)y^2$. It says that whenever $r^2 - as^2 = 1$, the map

$$(y, z) \mapsto (sz + ry, rz + asy)$$

preserves $z^2 - ay^2$. For $a = 3$, the smallest positive nontrivial solution of

$$r^2 - 3s^2 = 1$$

is $(r, s) = (2, 1)$. Substitution into the preceding map gives precisely

$$(y, z) \mapsto (2y + z, 3y + 2z),$$

which is the recurrence in (1). In the suggestive notation of quadratic norms, this is simply multiplication by $2 + \sqrt{3}$:

$$z_{n+1} + y_{n+1}\sqrt{3} = (z_n + y_n\sqrt{3})(2 + \sqrt{3}).$$

Since $(2 + \sqrt{3})(2 - \sqrt{3}) = 1$, this multiplication preserves the norm $z^2 - 3y^2$; the elementary calculation in the proof of Theorem 1 is exactly this norm argument written without invoking any algebraic-number-theoretic machinery.

Iterating gives the compact closed form

$$z_n + y_n\sqrt{3} = (6 + 2\sqrt{3})(2 + \sqrt{3})^n, \quad (8)$$

which follows at once by induction from the recurrence. Formula (8) explains both the rapid growth and the inexhaustibility of the family, while Theorem 1 supplies the completely elementary proof that every generated triple is integral, solves the original equation, and is distinct from all earlier ones.